Findings: experiments, confirmations and errors

Physicomimetics in BPP: what the conversion has found, kept up to date · 18 October 2026

A running record of what the conversion has found: the book's results we have reproduced, new results, errors and discrepancies in the book and in the other implementations, and our own mistakes, corrected. It is updated with every patch that finds something, and `FINDINGS.pdf` beside it is rebuilt each time. `BOOK.md` says what to build and `FIXES.md` what was fixed in the code; this file says what was learned.

How to read an entry. Each has an ID, the date, what was done, the numbers, where the evidence lives (a check script, a README or a probe), and a status: **Confirmed** (the book's claim, reproduced by our own measurement), **New** (a result the book does not have), **Book error** (the book disagrees with its own code, figures or arithmetic), **Book discrepancy** (the same, but small or an omission, not changing the conclusion), **Divergence** (NetLogo, the Android app and BPP differ), **Our error** (something we got wrong, and how it was found and corrected) or **Open** (measured but not yet settled).

Page numbers are PDF pages of `apbook1.pdf`; book page = PDF − 31.

1. The book's results, reproduced

F1. Appendix A: glibc `random()` is biased, and BPP uses it — 2026-10-17, Confirmed. `checks/rng_appendix.lua`. BPP's `math.random` is glibc `rand()` = `random()`, in Lua 5.1 and 5.2 alike (William's BPP links 5.2, the cloud build 5.1). Run with Listing A.1's sampling (S extra calls per accepted point in the unit disk, 10^6 points, 36 bins, 99.9% cut χ^2 66.6), it fails at S = 25 (χ^2 321) and 27 (409), as the book shows (PDF 646–648); Lua 5.2 the same (321, 410); also S = 26 (223) and 3 (70). The multiply-with-carry generator now in `lib/rng.lua` passes at every S tried (χ^2 26–47).

F2. Ch. 2.1 spring1D, every result — 2026-10-17, Confirmed. `checks/ch02.lua`, `ch02/README.md`. Momentum exactly 0; the separation oscillates symmetrically about D (27.000 to 73.000); period ×√10 at k = 1 (1.405 and 4.443); the centre of mass sits by the heavy mass at ratio 10; friction 0.001 drains the energy (2641 to 0.016) and the separation settles at D; changing D mid-run moves the energy to a new constant level (2641 to 998).

F3. Ch. 2.2 spring2D: angular momentum and the centrifugal stretch — 2026-10-17, Confirmed. `checks/ch02.lua`. Angular momentum constant to 2×10^{-14}, linear momentum 10^{-12}; mean separation 77.8 > D = 50; the maximum exceeds 2D = 100 for every seed tried (111–118), with spring2D.nlogo's random start (F41).

F4. Ch. 2.3 solar_system, experiments a–e — 2026-10-17, Confirmed. `checks/ch02.lua`. (a) A lighter sun gives a larger orbit, more so for AM 15 (87 to 133) than AM 10. (b) At Δt 0.1, AM 5 escapes (E from −121 to +111) and AM 0 is nonsense (E to 448,804). (c) At Δt 0.001, AM 5 orbits properly and AM 0 still shoots off left (x = −36,312). (d) Velocity caps fail: Vmax 9 drops the planet into the sun (|P| up to 891). (e) Force cap Fmax 1: bound, precessing rosettes (AM 4 precesses 112° per orbit); AM 0 swings through the sun, 0 to 15.6 away.

F5. Ch. 2.3e: the book's peak speeds are right — 2026-10-17, Confirmed. `checks/ch02.lua`. The text's "5.6, 5, 4 and 9 (when AM is zero, one, four and nine)" (PDF 84–85) looked non-monotone, and `book/ch02.md` advised re-measuring. We get 5.50, 4.97, 4.00, 9.00. For AM 4 and 9 the start is the closest point, so the peak is the starting speed; AM 0 falls straight through the sun and AM 1 nearly does, so both peak near the centre.

F6. Ch. 11.3: the mean free path 0.651757, and 0.6079 with a buffer — 2026-10-17, Confirmed. `ch11/README.md`, `checks/mfp_theory.lua`. The integral, computed to 60 digits, is 0.6517567914559… (Eq. 11.4 gives 0.651757); the 5% buffer gives 0.607939 (PDF 383: 0.6079). Listing 11.4 at 10^7 samples agrees within one standard error.

F7. Ch. 11.2: coverage converges to a seed-independent limit — 2026-10-17, Confirmed. `HANDOFF.md` §6. As the book argues from ergodicity (PDF 380), long runs from different seeds reach the same deviation: at MFP 50 / 100 with one agent, 0.0158 / 0.0368 from both seeds to within 1%.

F8. Fig. 4.19: 100 agents cover more evenly than 1 at large mean free paths — 2026-10-17, Confirmed. At equal visits (5×10^7), MFP 30 / 50 give 0.0035 / 0.0158 with one agent and 0.00087 / 0.0035 with 100. Crowding adds turns, shortening the effective free path. Agents are near-independent only when they rarely meet.

F43. Ch. 13.2: the split-Newtonian point mass dithers across D — 2026-10-18, Confirmed (E2, PDF 447–449). `checks/ch13.lua`, `probes/settle_drones.lua`. The book says the sign flip at D lets a point mass dither across it with almost no net motion. The 2D particle replica of the drone law (G 1500, power 2.2, forceMax 1.5, D 50, friction 0.6, 40 particles) does exactly that: after 1,500 ticks the bonds are at 0.99 D but every particle still moves 0.55 units per tick. Flown by 50 drones the dither becomes real motion that never stops: 0.27 m/s (2D) and 0.47 m/s (3D) after 170 s, 190 and 324 AR-boundary crossings per drone per minute, and a suggested-position gap of 6 and 11 cm (3 seeds each). Lennard-Jones on the same drones settles in 20 s (2D) and at the last launch wave (3D).

F44. Ch. 13.5: AR damping η settles the Newtonian drones like LJ — 2026-10-18, Confirmed on drones (the book's Pioneers settle, PDF 454). `quadcopter/README.md` ("Ch. 13 on the drones"). With η (Eq. 13.9, power 1) and nothing else, the same 50 drones settle (scalar acceleration below 0.1 m/s² from then on) 6 s after AP engages in 2D and 29 s in 3D, when the last launch wave joins (LJ: 20 s and 31–33 s); AR crossings fall to 0–1 per drone per minute, speed to 0.001–0.003 m/s, the gap to 0.03–0.09 mm, with no crashes (baseline 2 per 2D run). This answers `BOOK.md` §1 item 1's expectation: yes, η alone does it, because the law is then smooth through zero at D. The square and cube formations settle too: 6 s (2D square, LJ 23–25 s) and 27 s (cube, LJ 33 s), against 0.33 and 0.63 m/s of endless jitter without η; LJ's cube is the more regular (bond CV 0.007–0.03 against η's 0.07–0.09). The lattice η settles into is the subject of F45.

2. New results

F9. The mean free path in 3D: 0.7720 × the cube edge — 2026-10-17, New. `ch11/README.md`, `ch11/mfp_exact.py`, `checks/mfp_theory.lua`. For cubic cells with face neighbours, the mean distance from a robot uniform in a cube to entering the next is 0.771991712645..., or 0.713804 with the 5% buffer. An integer-relation search on the 60-digit integral finds $90\,f = 4 + 17\sqrt{2} - 6\sqrt{3} - 7\pi + 21\ln(1+\sqrt{2}) + 42\ln(2+\sqrt{3})$, agreeing to 50 digits (found by search, not proved). Listing 11.4 extended to 3D agrees within one standard error. In BPP's coverage box (edge 26⅔) that is a mean free path of 20.6, or 19.0 with the buffer, against 17⅓ in 2D.

F10. The 2D mean free path in one line — 2026-10-17, New. $12\,f = 2 + \sqrt{2} + 5\ln(1+\sqrt{2})$, equal to the book's Eq. 11.3–11.4.

F11. Uniform Coverage: deviation limits by agent count — 2026-10-17, New. At 5×10^7 visits (ticks × agents), mean of two seeds:

MFP	1 agent	10 agents	100 agents
1	0.0046	0.0036	0.0031
4	0.0028	0.0020	0.0024
10	0.0019	0.0023	0.0014
17	0.0021	0.0020	0.0014
30	0.0035	0.0036	0.00087
50	0.0158	0.0131	0.0035
100	0.0368	0.0347	—

MFP 30–100 have converged (seeds agree); MFP ≤ 17 are still falling as 1/√visits and are upper bounds. At equal visits one agent covers as evenly as ten, so what agents buy is time (F8 is the exception, when they crowd).

F12. The book's integrator: energy swings but does not drift — 2026-10-17, New. `checks/ch02.lua`, `checks/monitors.lua`. Semi-implicit Euler keeps energy bounded, with a swing inside each cycle of about $\Delta t\cdot\omega$: 0.45% for spring1D at Δt 0.001, 16% at the closest approach of the AM 15 orbit at Δt 0.1, 8% for AM 5 passing 1.6 from the sun at Δt 0.001. The drift between the start and end of a run is 10^{-4} or less for the springs and 0.2% for the

coarsest orbit. The energy error is first order in Δt (10.2× smaller for 10× smaller Δt). "Constant energy" in the book means no drift.

F13. Ch. 3's obstacle field, measured from the figures — 2026-10-17, New. `lib/world.lua` (`World.CH03_DISCS`). Both panels of Fig. 3.25 give the same 12 disc centres within a pixel, shifted as a group (F40). Scale 2.11 px per patch at 300 dpi from the discs (20 across), confirmed by the formation's 63.3 px node spacing (D = 30). In the model's world (701 × 401, goal at (−175.25, 0); F40) the field's centre is at (−8, −11); the closest gap is 5.2 patches ("narrowly spaced"). Fig. 3.26 is a different field (F37).

F42. spring3D stays in one plane — 2026-10-18, New. `ch02/Spring3D.lua`: from random 3D starts, L keeps its direction to 2×10^{-12} degrees.

F45. η shrinks the lattice when the range reaches second neighbours — 2026-10-18, New. `checks/ch13.lua`, `quadcopter/README.md`. In 2D at the default range 1.5 D, η settles the drones at 0.83 D (power 1) and 0.805 D (power 2.2) instead of D, on all 3 seeds; the particle replica gives 0.82–0.85 and 0.79–0.81, so this is the law, not the drones. η damps the nearest neighbours' repulsion near D far more than the attraction of the second neighbours, which come inside the range (at √3 a) once the lattice spacing a falls below 0.866 D. Balancing the two in an infinite triangular lattice gives a / D = (1 + c) / (1 + √3 c), with c = 3^((1 − P) / (2 p_η)) for law power P: 0.800 (p_η = 1) and 0.762 (p_η = 2.2); a finite swarm's edge, with fewer second neighbours, sits a little wider, as measured. Keeping the second neighbours out restores the spacing: range 1.25 D gives 0.986 D on the drones (1.001 D as particles), with the best lattice of any run (Delaunay CV 0.069 against LJ's 0.092 and the baseline's 0.11). The 3D triangle's default range is already 1.25 D, and it settles at 1.01 D.

F46. The stop rule needs η — 2026-10-18, New. With the sign flip, the summed force is almost never small: a stop threshold of 0.01 or 0.03 (AP units; in 2D the force just either side of D is ±0.27 per neighbour) held 0.04–0.3% of drone-ticks in 2D and none in 3D, and the swarms jittered on as before. With η it holds every drone once settled (residual speed 10^{-4} m/s in 2D, 10^{-9} m/s in 3D), with the same settling time and lattice as η alone, so it does not freeze the swarm before it has formed.

F47. η softens the lattice: smooth flight, but it gives under load — 2026-10-18, New. With η (power 1) the force's slope at D is F(D)/D: 0.0055 per AP unit in 2D and 0.0085 in 3D, against LJ's $c{\cdot}p/D^2 = 0.024$, so the lattice is a quarter to a third as stiff (the plain law has no slope at D, only a step). Under the same load η therefore needs about four times LJ's strain. Crossing the default obstacle field after the moving goal (640 s, 2 seeds), η's drones fly smoothly (acceleration below 0.1 m/s² throughout, mean speed 0.10 m/s against the jittering baseline's 0.20 / 0.40 m/s), lose none, and travel slightly farther than the baseline but less far than LJ (swarm centre at 632 s: −46 m against −43 and −50 in 2D; −47 against −41 and −51 in 3D). Under the goal's constant pull their bonds shrink to 0.85–0.92 D in 2D and 0.94–0.95 D in 3D, where LJ's stay at 0.97–1.0. η at the law's own power (2.2 in 2D) settles no better than power 1 and leaves the lattice smaller (0.87 D at range 1.25, still creeping outward after 170 s), so power 1 is the better default.

F49. η has two resting states: switched on after forming, the lattice stays at D — 2026-10-18, New; found by William in the GUI. With η on from the launch, the 2D swarm settles at 0.83 D (F45). Switched on once the plain law has spread the swarm (60 s after AP engages, 3 seeds), the same drones settle within about 2 s at 0.96–0.97 D in 2D, and at 1.00–1.01 D in 3D with bond CV 0.035–0.055, the most regular Newtonian lattice measured (η from the start 0.063, LJ 0.044, plain law 0.083). The particle replica does the same (0.988 D; `checks/ch13.lua`). Both states are equilibria: from a compact cluster the expansion passes through spacings below 0.866 D, where second neighbours are in range and η's balance stalls it; a lattice already near D has them out of range, and η only removes the dither. William's reading: it is annealing, with η as the cooling — the plain law's dither (Ch. 13's "jumpiness", PDF 449) is the heat that lets the lattice reach D, η from the start is a quench into the compressed state. Next: a cooling schedule (ramp `etaPower` 0 → 1) and the fastest cooling that still reaches D.

F50. η in a band tied to D: no trap, a stiffer lattice, and a stiffness limit set by the drones — 2026-10-18, New; William's idea. Damping only within $|D - r| < w{\cdot}D$ (`etaWidth`; w = 1 is Eq. 13.9) leaves the far forces alone, so from the launch the 2D swarm no longer stalls: bonds 0.83 D (w 1), 0.85 (0.5), 0.98 (0.25), 0.999 (0.15) on 50 drones, 3 seeds each; the particle replica reaches D for w ≤ 0.2, as the F45 balance predicts (no trap for w below about 0.26).

w 0.15 gives the best 2D lattice measured (Delaunay CV 0.064; LJ 0.092). Too narrow brings the jitter back: 2D at w 0.1 crashed 4 drones, and 3D at w 0.15 kept jittering in 2 of 3 runs, at 0.1 fully (0.32 m/s). The edge falls at the same slope of the damped force at D, F(D)/(w·D), in both dimensions: 0.034–0.037 per AP unit settles, 0.055–0.057 does not (LJ's slope is 0.024). So the band can be set from the law and the separation, w ≈ F(D)/(0.04 D): 0.14 in 2D, 0.21 in 3D. One w = 0.2 works in both: 2D settles in ~10 s at 0.994 D (Delaunay CV 0.077), 3D at the last launch wave at 1.01 D (bond CV 0.054), no crashes. Crossing the obstacle field after the goal (2 seeds), w 0.2 holds its bonds at 0.93–0.94 D (2D) and 1.00 D (3D), where w 1 gave 0.85–0.86 and 0.94–0.95 and LJ 0.97 and 0.99, but it is slower: swarm centre at 632 s −43 m in 2D and 3D (w 1: −46, −47; plain law −43, −41; LJ −50, −51). The stiffer lattice threads the columns no faster than the jittering one; LJ is both stiff and fluid.

3. Errors and discrepancies in the book

F14. Listing 11.4's printed result — 2026-10-17, Book discrepancy. `ch11/listing_11_4.lua`, `ch11/README.md`. The listing's entry is at $x_2 = 1.001$ (exact mean 0.652501); the text prints 0.6517. With the book's generator (glibc, srandom(1), Lua 5.1) the listing gives 0.652454, 9 standard errors away, but 0.651711 = "0.6517" at $x_2 = 1$, so the text's run most likely used $x_2 = 1$ (inferred).

F15. Ch. 2.3d: the velocity cap's effect on the orbit's shape — 2026-10-17, Book discrepancy. `checks/ch02.lua`. At Vmax 12 the book says the orbit "runs well, with a slight linear-momentum change". It is slight ($|P| \le 3.0$), but the orbit shrinks unmentioned from 15–87 to 15–18: the cap cuts the start speed 15 to 12. solar_system.nlogo caps as we do (scaling the speed, keeping the direction); clipping each component gives 14.9–18.4.

The chapter notes list further suspected errors in the listings (Ch. 3, 6, 7, 9, 13, 18; `BOOK.md` §1 item 11). They are not yet re-checked by measurement and will be entered here as each chapter is ported.

4. Differences between implementations

F16. Uniform Coverage: the wall rule decides the large-MFP results — 2026-10-17, Divergence (kept by decision). `FIXES.md` 36. NetLogo turns when the patch 1 ahead or either patch 2 away at ±45° is wall; the Android app, and BPP from it, turns only on touching the boundary. With NetLogo's rule swapped into BPP (one agent, 10^7 ticks), MFP 30 / 50 / 100 give 0.0074 / 0.0227 / 0.0271, reproducing NetLogo's own 0.0072 / 0.023 / 0.026. The Android rule gives 0.0035 / 0.0158 / 0.0368: better at every useful MFP, worse only at 100. NetLogo is not maintained; the Android/BPP rule stays.

F17. Uniform Coverage: the two known Android bugs barely matter — 2026-10-17, Divergence. `FIXES.md` 4–5. The one-sided neighbour cone (0–30° for ±15°) and the dropped ninth move were real, but move the deviation about 3%, so BPP and Android agree (MFP 30, 100 agents: 0.00087 against 0.00085).

F18. Earlier NetLogo and Android numbers at small MFP came from longer runs — 2026-10-17, Open. William's old values at MFP ≤ 17 (0.00027–0.0016) sit below our 5×10^7-visit upper bounds, so those runs were longer. Their lengths were not recorded.

F19. Zarzhitsky's map format, kept by the Android port — 2026-10-17, Confirmed. `lib/world.lua` (`writeCPT`). A text file of nx lines of ny characters, `' '` open, `'b'` blocked. His PlumeGen, rebuilt from source, loads a map written by our generator and reports the same coverage (3.6%). Its reader as shipped wants a square of at most 1024; ours writes any size.

F48. `formation_newton.nlogo`, the book and the BPP port use different defaults — 2026-10-18, Divergence. The law is the same in all three (`netlogo/README.md`). The model's sliders are G 310, power 2, Force_Maximum 1, Desired_Separation 30, Friction 0.15, Time_Step 1, 50 particles; the book's text gives G 400 for the same model (`book/ch03.md`); both are above the clustering threshold G_t = 259.8, so the model clusters by design. The BPP ports take their 2D defaults from the Android app's `MyPanel.java`: G 1500, power 2.2, forceMax 1.5, D 50, friction 0.6, which is below its G_t (2368). Ch. 13's damping is in none of them.

5. Our own errors, corrected

F20. "NetLogo's MFP 30 and 50 values come from short runs" — 2026-10-17, Our error. With NetLogo's wall rule BPP reproduces NetLogo's values (F16), so they had converged; a short run can only raise the deviation.

F21. The seed slider would not move — 2026-10-17, Our error. `FIXES.md` 34. Found by William. BPP stores slider ranges as floats and turns them into an integer slider; a maximum of $2^{31} - 1$ overflowed it. Now 0–1000; larger seeds go through `SEED`.

F22. R in Uniform Coverage reset the seed — 2026-10-17, Our error. `FIXES.md` 34. R was BPP's own reload, which resets every slider to its default, so a chosen seed never applied. Uniform Coverage now binds R to a new scatter, and a seed change restarts the scene in every demo.

F23. The world viewer was mirrored top to bottom — 2026-10-17, Our error. Found by comparing with Fig. 3.25. Book y now maps to BPP −z, so a camera looking down −z sees the book's orientation.

F24. Energy "flat" tested as no swing — 2026-10-17, Our error. The first Ch. 2 checks failed seven times on the integrator's normal within-cycle swing (F12); they now test drift and report the swing separately.

F25. The repository ignored `lib/` — 2026-10-17, Our error. The root `.gitignore` rule `lib` (for build output) silently left the new folder out of the first commit; an exception was added.

F26. A test hook replaced the scene's own — 2026-10-17, Our error. BPP keeps one `postSim` callback, so a probe that set its own stopped the simulation. Probes now use `preDraw` and `preStop` (`lib/batch.lua`).

F27. The 3D mean free path estimate — 2026-10-17, refined: an earlier ≈ 0.7717 (4 M samples) is within error of the exact 0.771992 (F9).

F33. The checks assumed `lua5.1` was installed — 2026-10-17, Our error. Found by William (Lua 5.3 only): every check failed for want of the interpreter, and three used `math.atan2`, gone in 5.3. `checks/run_all.sh` now runs them inside BPP, else any Lua; all code runs on 5.1–5.4.

F37. Fig. 3.26 drawn as Fig. 3.25's discs enlarged — Our error, found by William. Obstacles are placed by mouse clicks, so Fig. 3.26 has its own field: six discs fitted to its outline (preset `ch03-wall`); drawn over the figure, they trace it.

F38. Fig. 3.26's discs measure radius 45, the text says 50 — Book discrepancy (small). Radius 95 px at the scale confirmed by the lattice (F13); the field covers 24.8% of the panel, ours 24.6%.

F39. Code against William's instruction went out in a commit of ours — Our error. Another session's `cpt-textron` preset reached master because I staged the folder unread. Removed; diffs are now read first.

F40. Both figures' obstacles shift between panels: cropping — 2026-10-18, Our error (small), settled by the `.nlogo` and Fig. 3.23: each panel crops ~305 patches of a 701 × 401 world, which we took for the world. Pushing moves a heavy (10^5) obstacle ≤ 0.01 patch/tick. Presets use the model's frame.

F41. Ch. 2's starts guessed, then checked against the .nlogo — 2026-10-18, Our error (small). spring2D now starts at random headings and spring1D's light mass on the right, as the models; F3 barely moves.

6. How BPP behaves (worth knowing for every port)

F28. Headless results depend on `delayMs` (FIXES 1): its throttle uses wall-clock time; `lib/batch.py` sets it to 0.

F29. R reloads the script and resets every slider unless a scene binds R itself; the toolbar restart restores the sliders after reloading (F22).

F30. One callback of each kind: `v:postSim`, `v:preDraw`, `v:preStop`, `v:onParamChanged`; setting another replaces it (F26).

F31. Slider ranges are floats turned into integers (value / step): keep them well inside $\pm 2^{31}$ steps; values above 2^{24} are not exact (F21).

F32. Scripts run in their own folder in the GUI; scripts in subfolders add `../?.lua` to reach `lib/`.

F34. BPP's Lua depends on the build: 5.1 or 5.2 (`bpp.pro`), not the standalone `lua`: William's is 5.2, the cloud build 5.1 (`bpp -l 'print(_VERSION)' -n 1`). The checks and the generator's stream agree on 5.1–5.4.

F35. BPP runs a scene as soon as it loads; S stops and restarts it. GUI tests that "pressed S to start" were stopping runs (our error).

F36. Huge "off" values make unusable sliders (Force_Maximum 0.1–10^6 could not be set to 1; found by William). Caps now get a tick box and a slider over the useful range. Same root as F31.

7. Open questions

- F18: the limits at MFP ≤ 17, which need about 10^9 visits to resolve.
- The 3D threshold G_t for the cube formation (`BOOK.md` §1 item 9), to measure with a G sweep.
- How to keep η's spacing at D in 2D at range 1.5 D (F45): a shorter range works, as do switching η on after the lattice forms (F49) and a narrow band tied to D (F50). Whether the slope limit (~0.04–0.05 per AP unit) follows from the drones' lag and AP rate, and holds for other apHz. And whether the next Ch. 13 mechanisms (adaptive and static friction) settle the drones without η's loss of stiffness (F47).